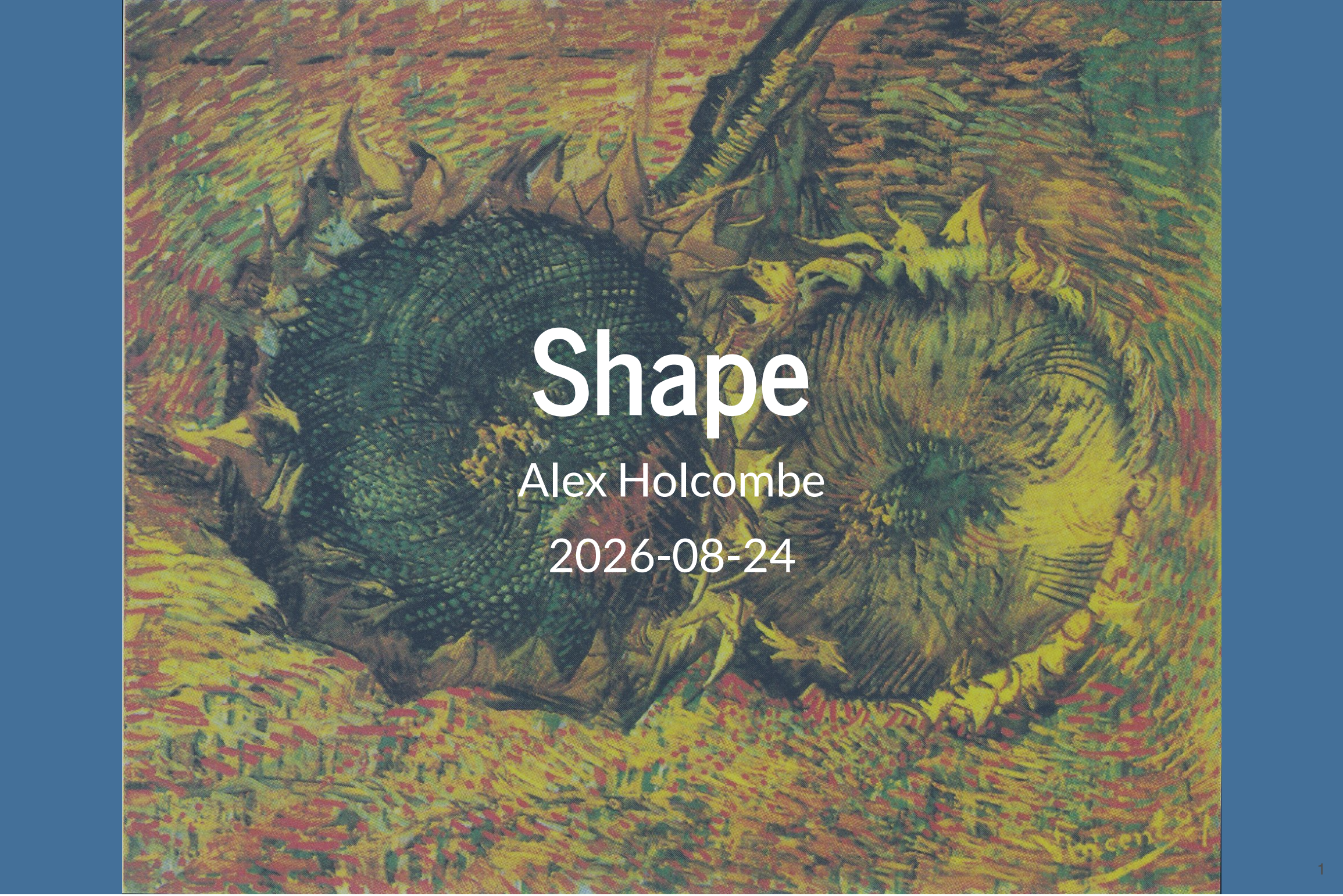




Shape

Alex Holcombe

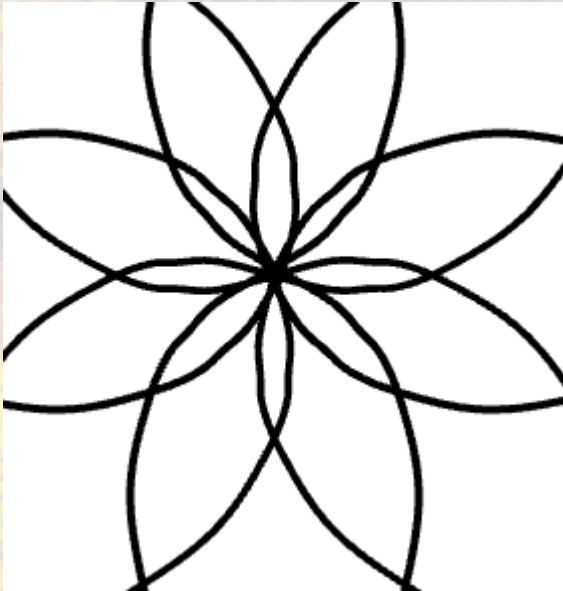
2026-08-24

What about pairing *within* a feature dimension?

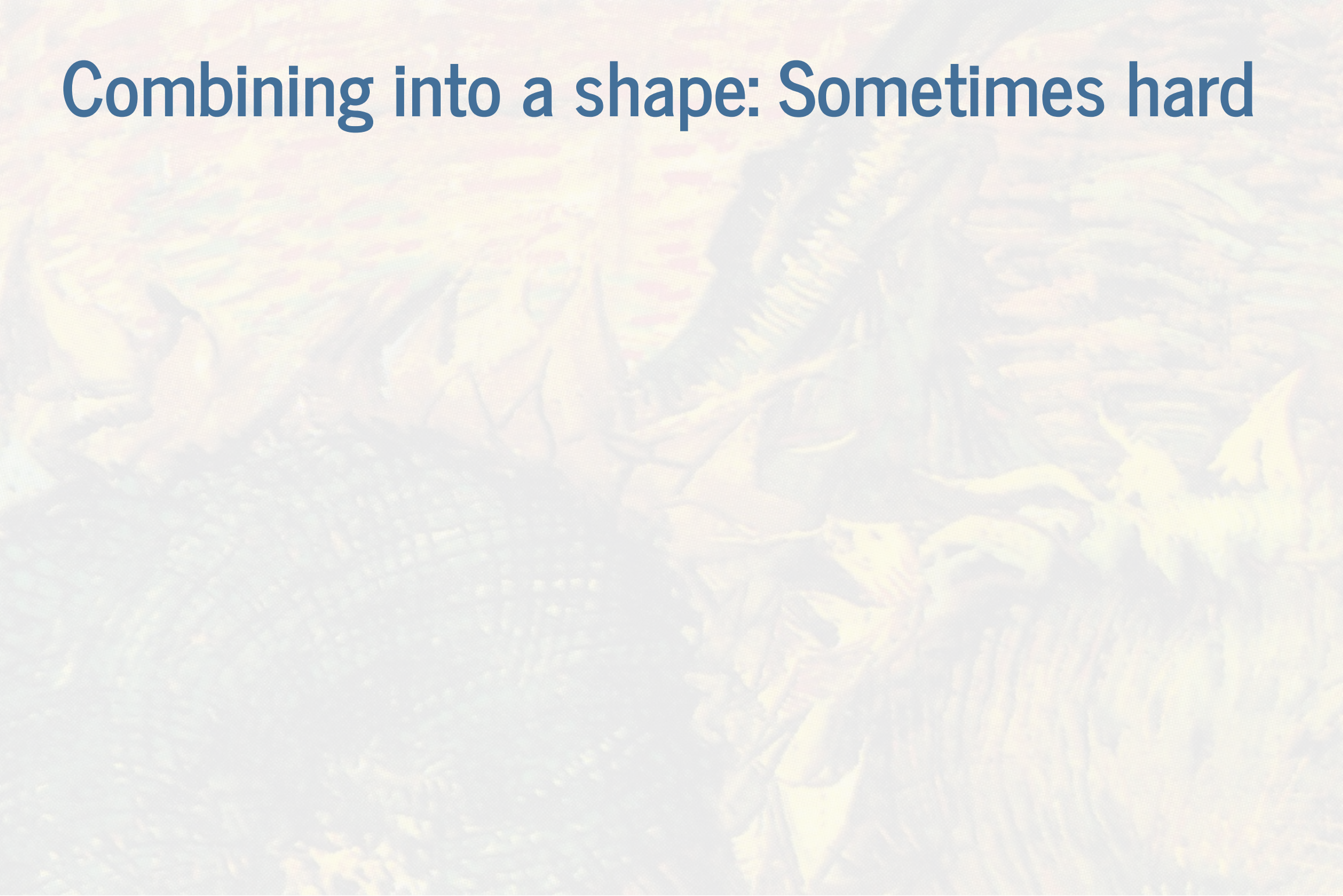
- Within-feature salience is computed pre-bottleneck (Chapter 5)
 - That's *differencing*
- What about combining parts into a larger shape?

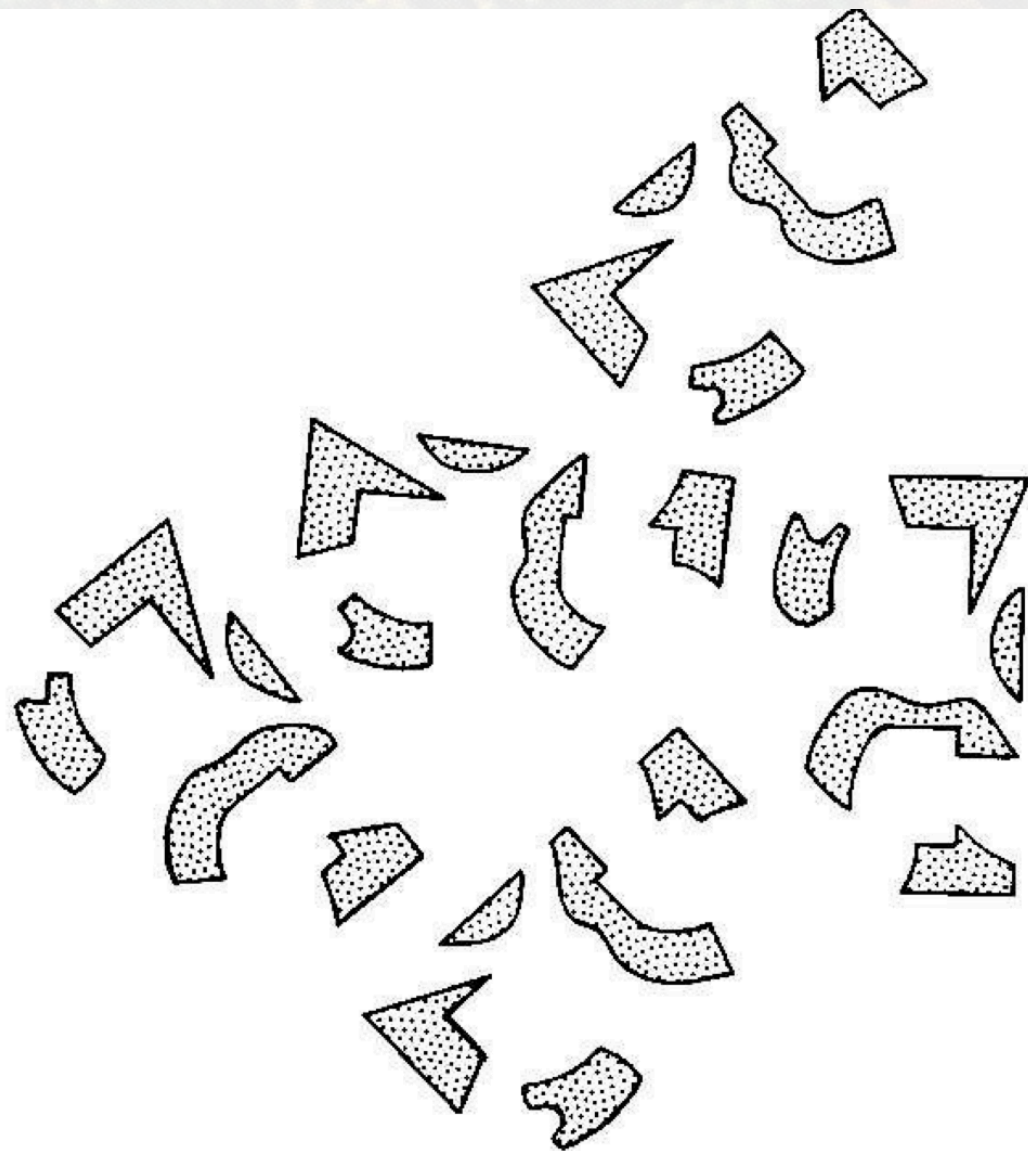
Shape

Involves combining elements of a single feature dimension:
orientation

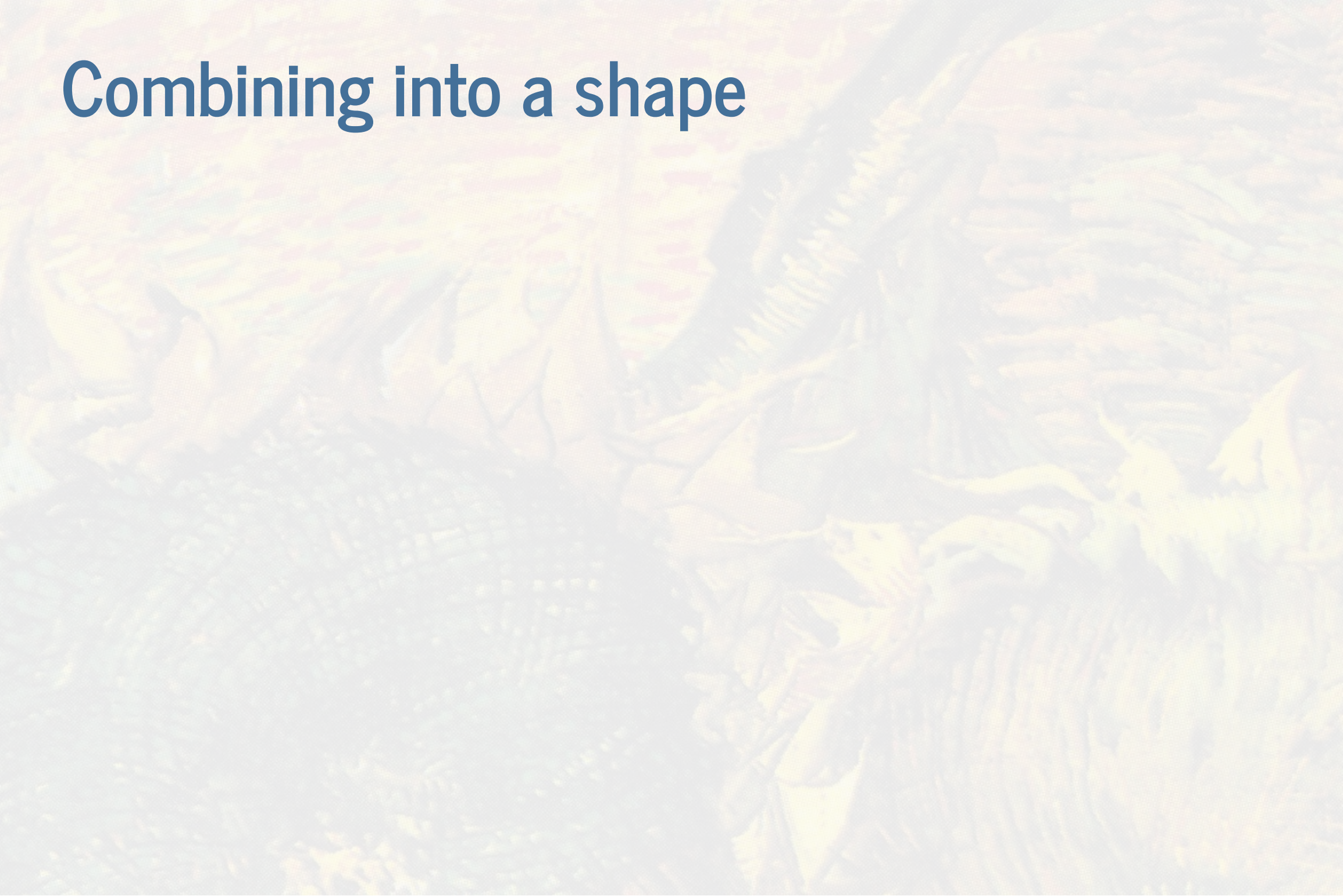


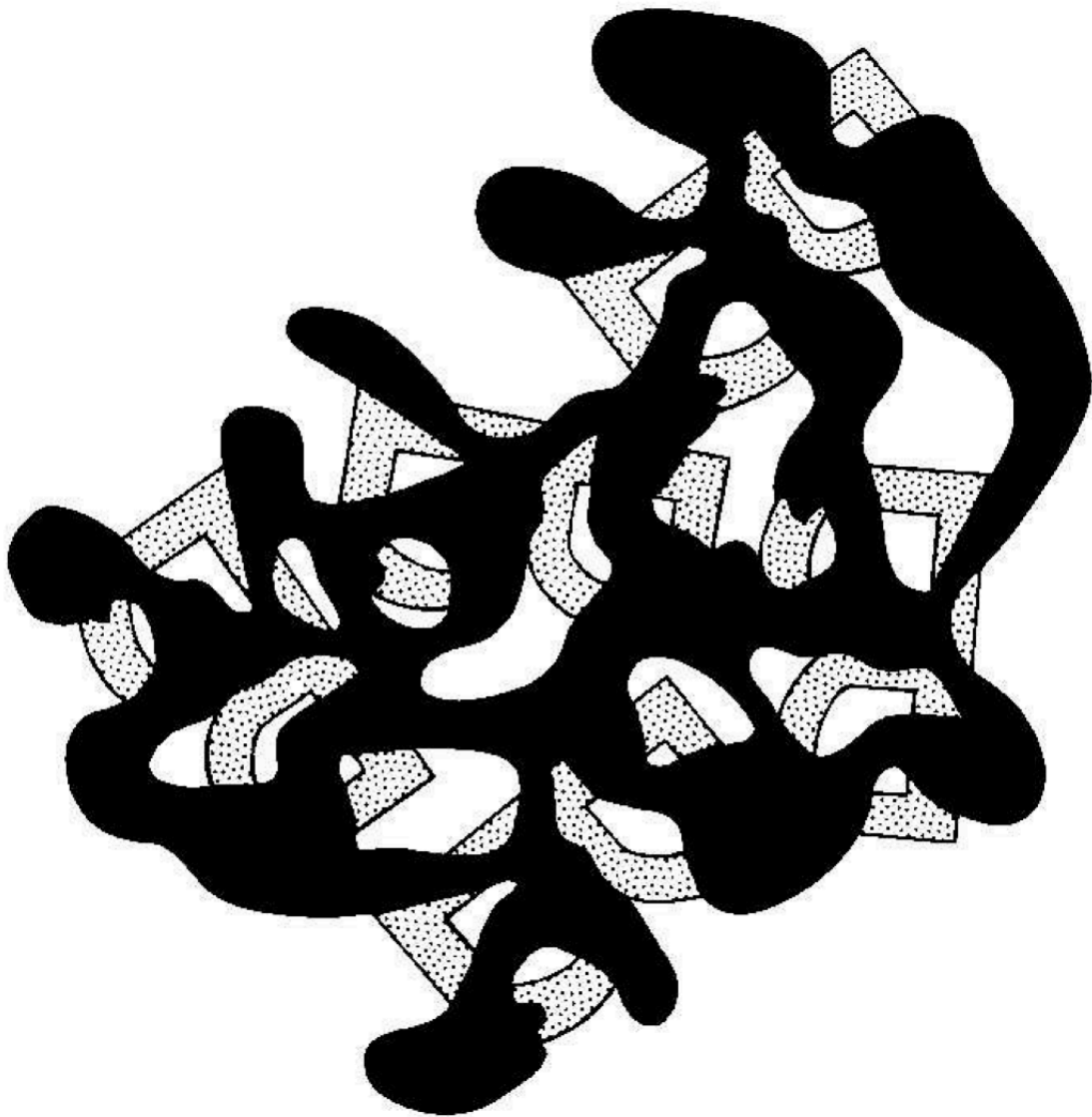
Combining into a shape: Sometimes hard





Combining into a shape



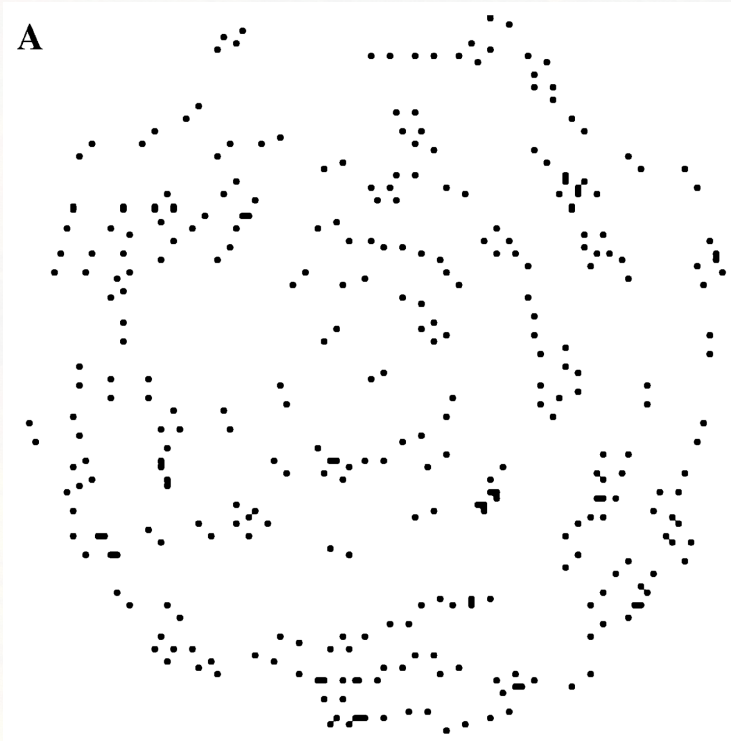


Shape processing

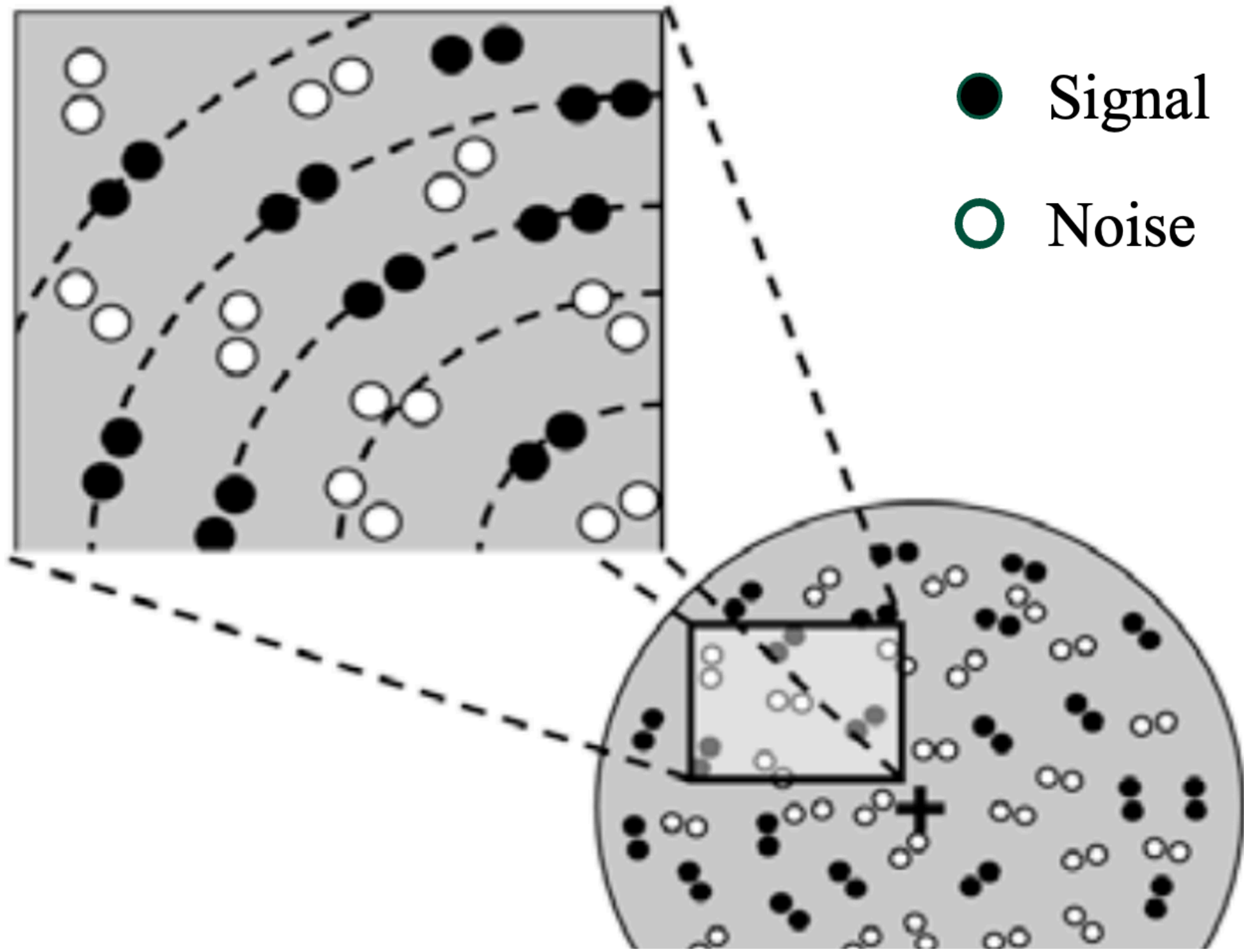
Combines elements in a sophisticated way.

- Occlusion cues
- Next: averaging out noise

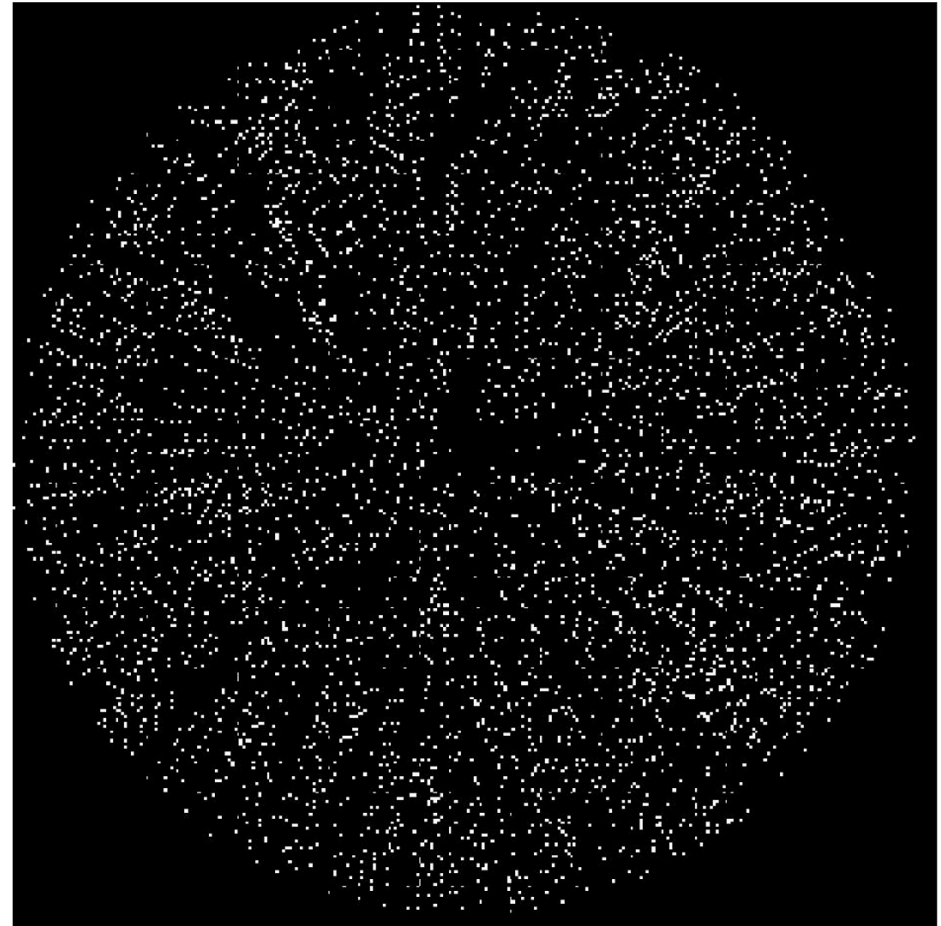
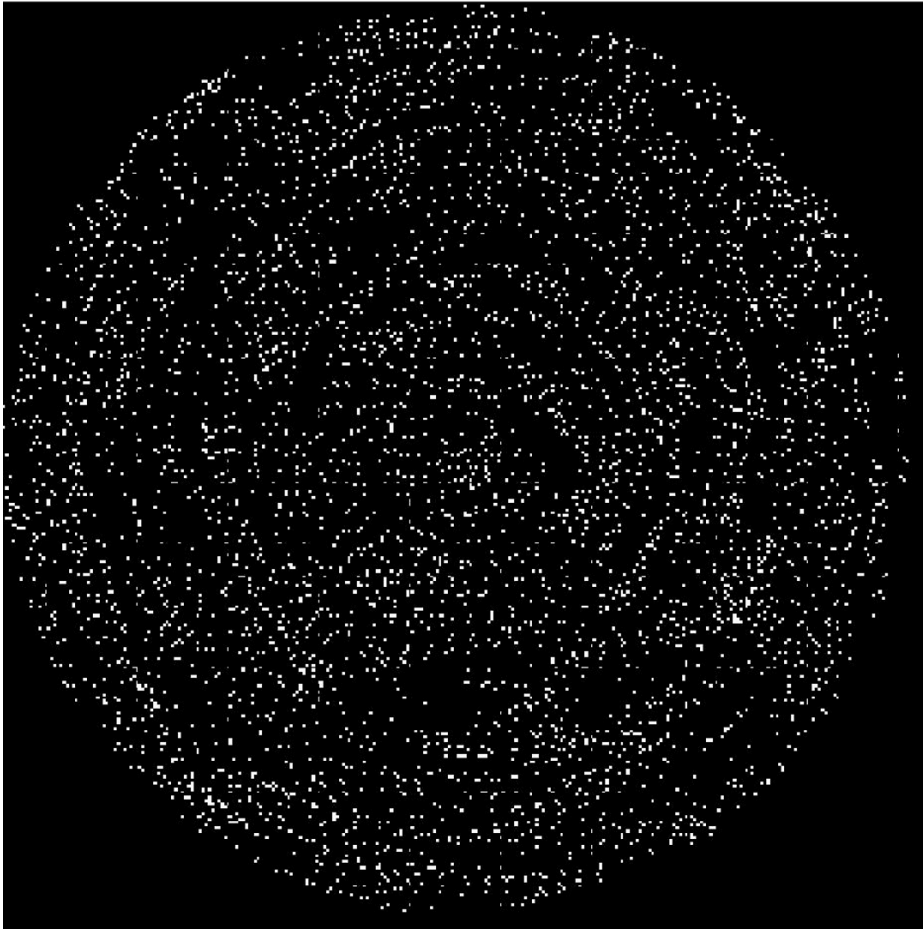
Quantitative assessment of combining



Dipole = two dots, defines an orientation



Circular or radial?



Circular or radial?



Circular or radial?

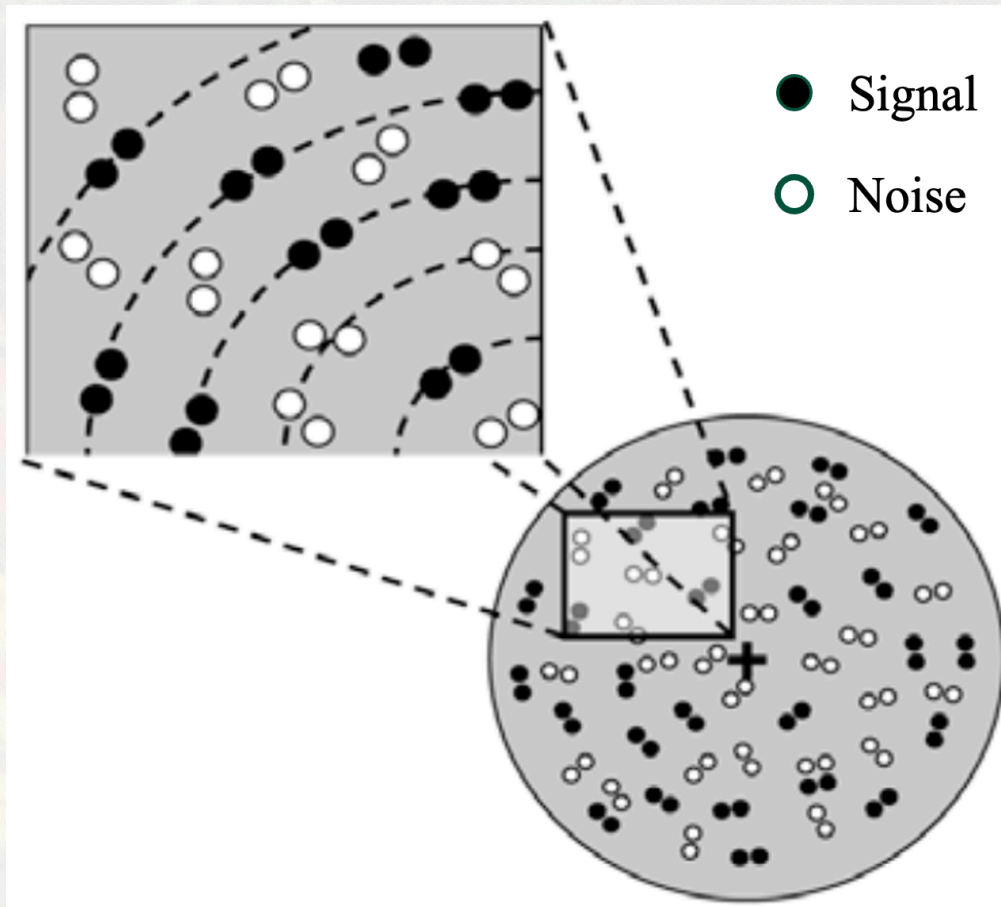
Circular or radial?



Global shape processing?

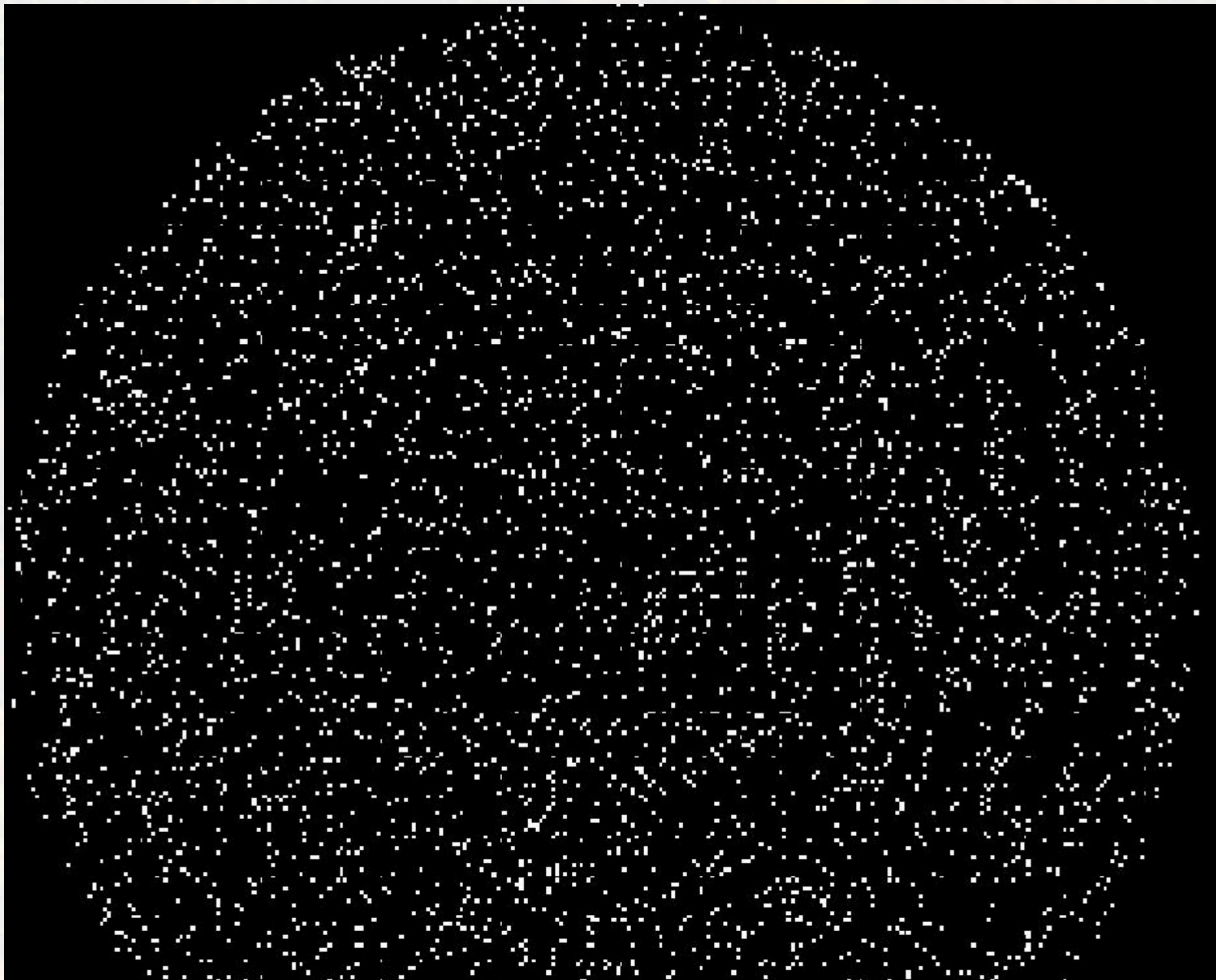


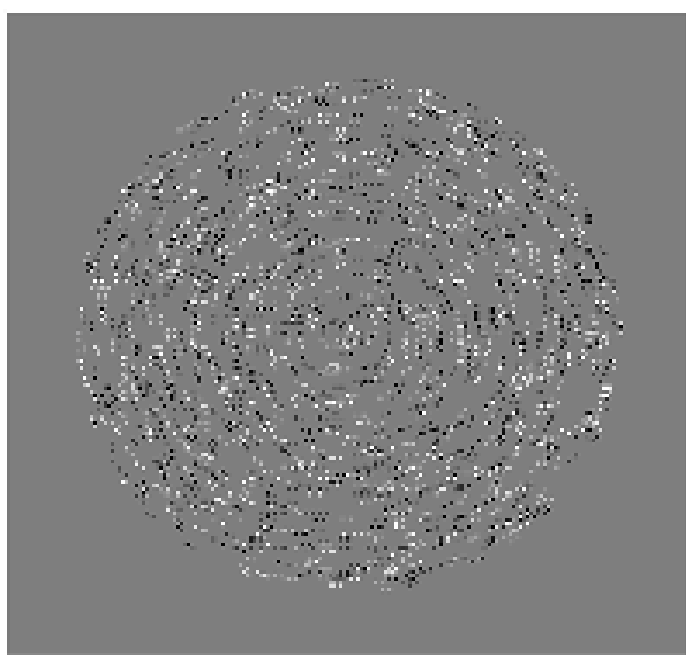
- In previous examples, answer can be determined by combining just 4 dots
- Add random dots (noise)
 - Can only get answer if combine lots



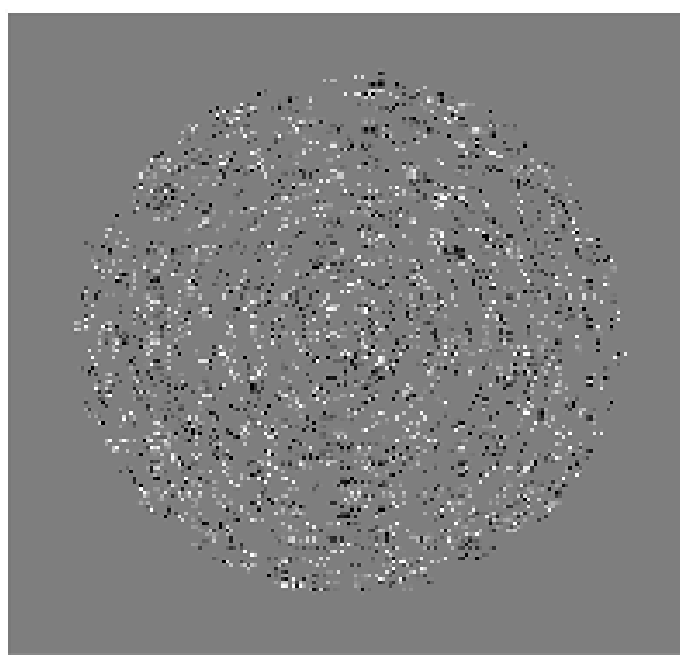
40% of dots random



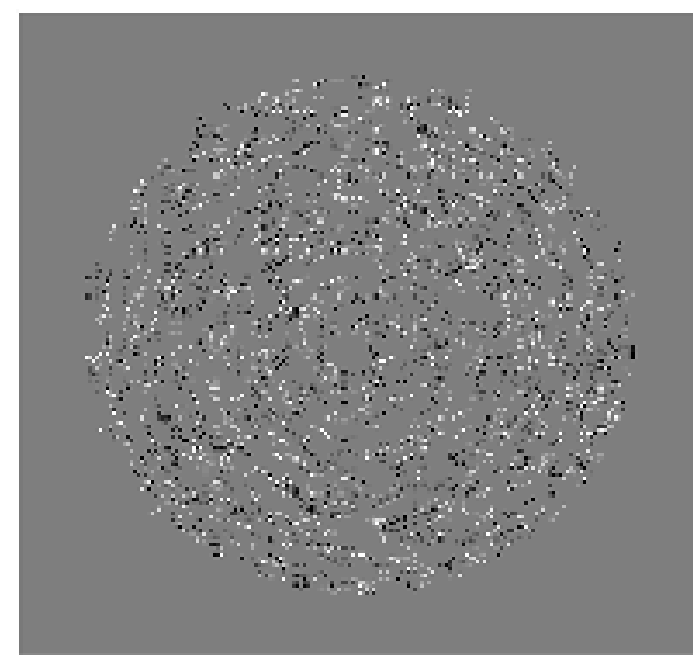




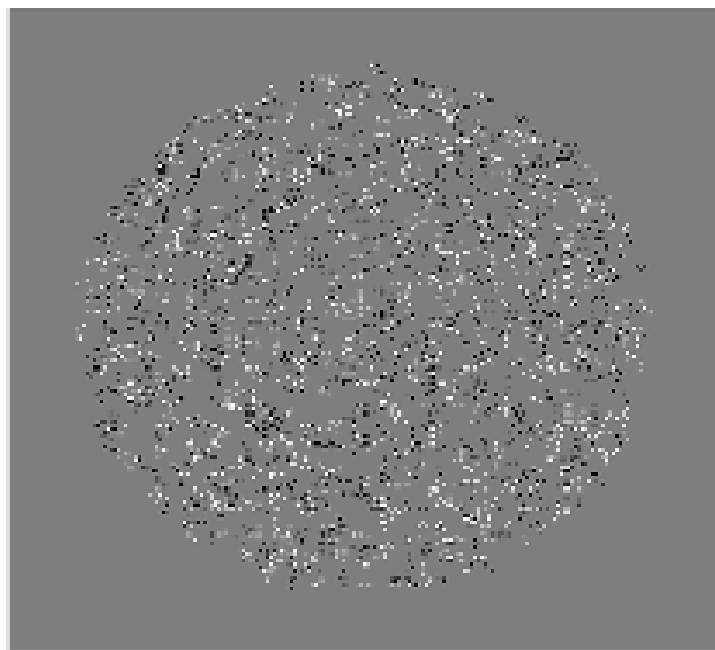
0% noise



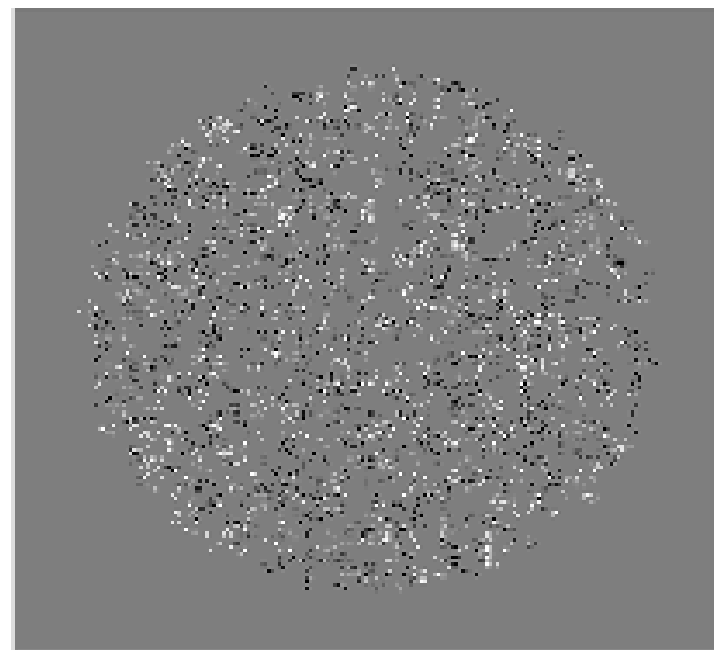
25%



50%



75%



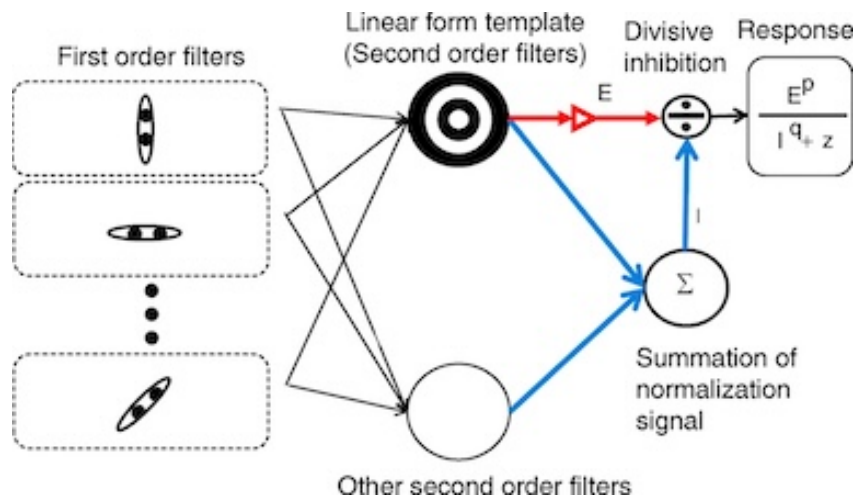
100%

Radial



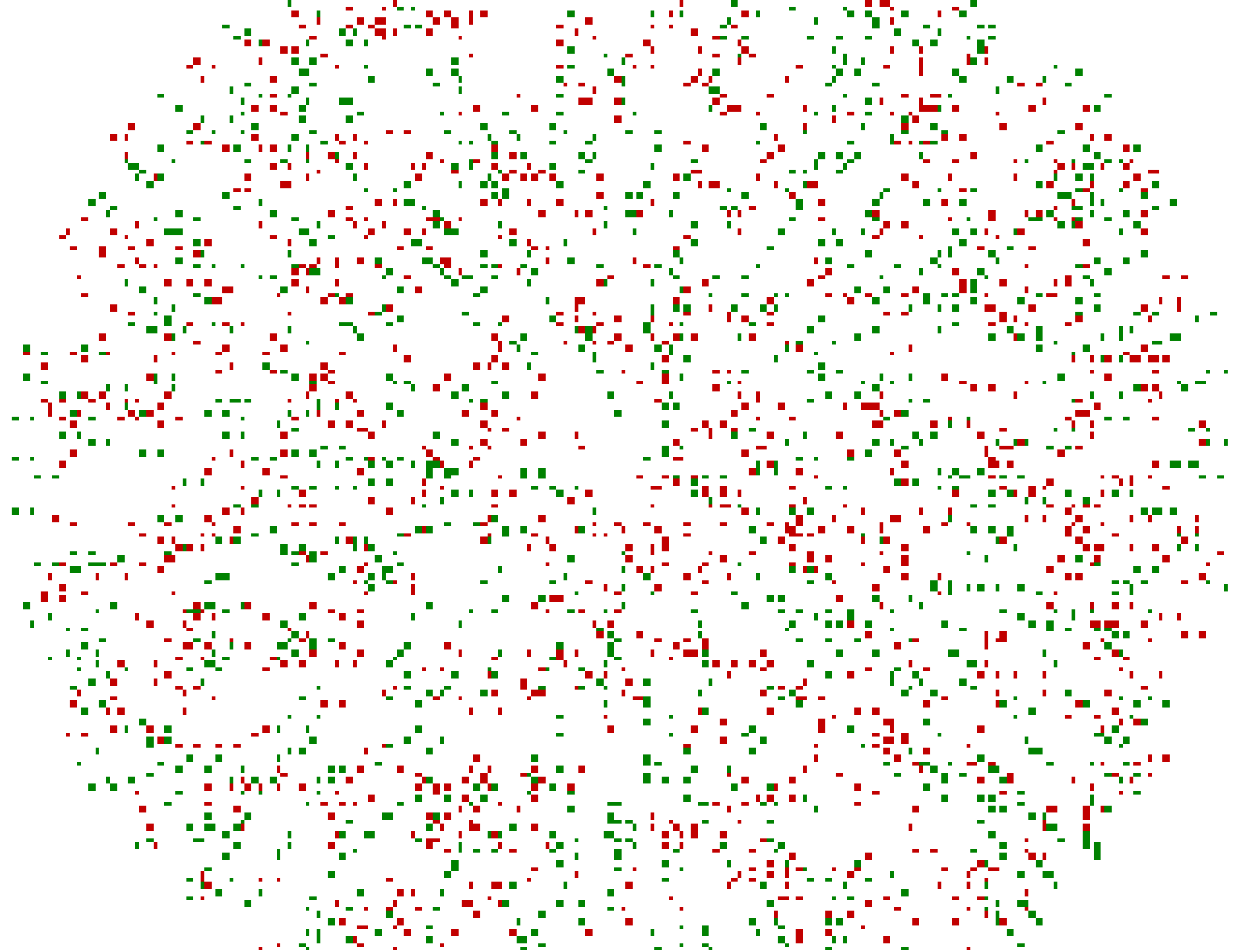
Theory

Simplest possible theory consistent with behavior and neurons

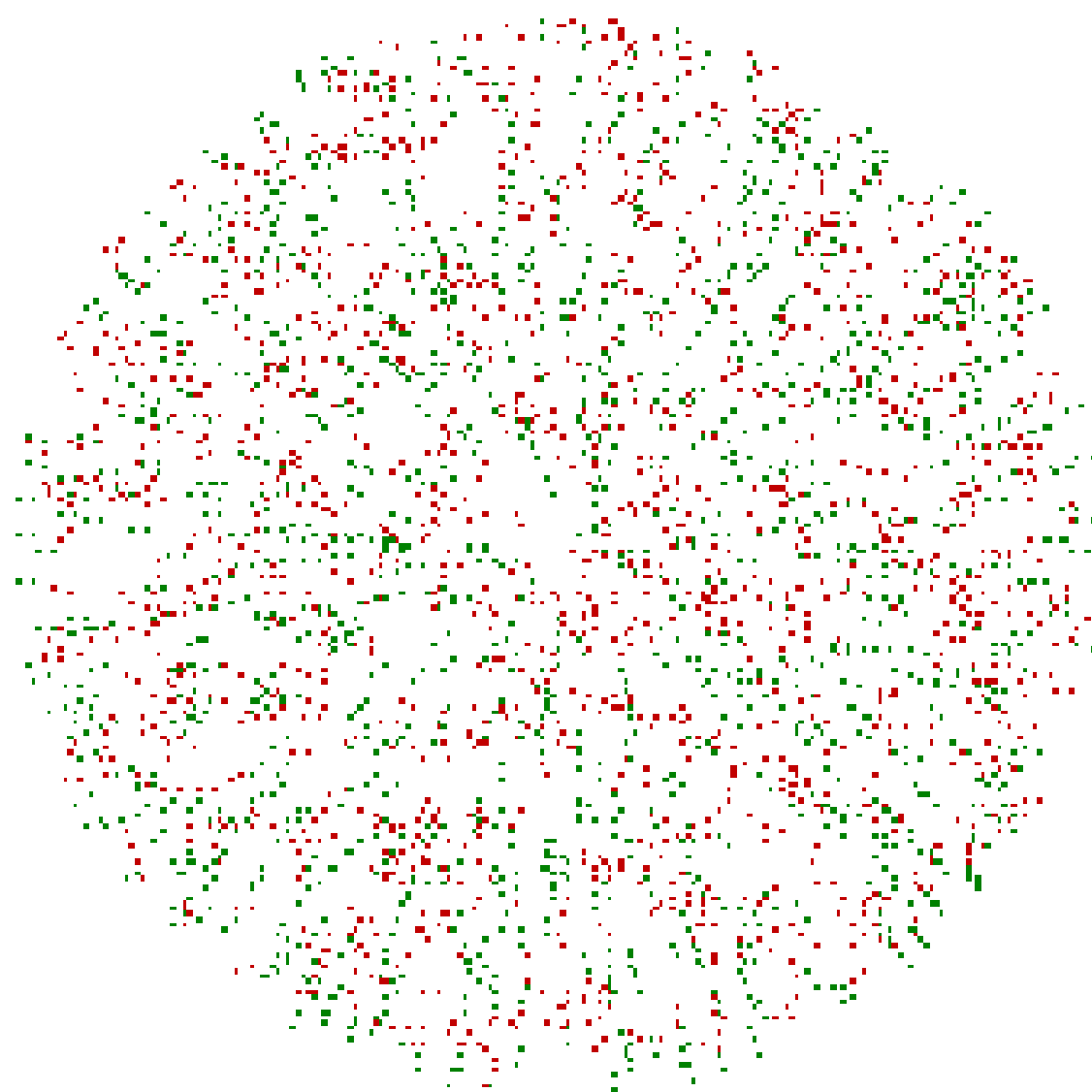


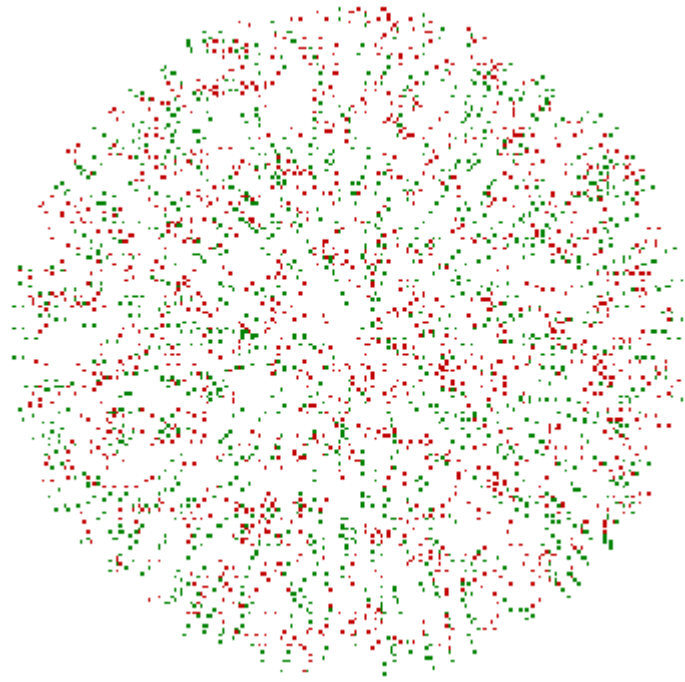
- Can't explain other aspects of shape perception

Radial or circular?



- What color were the radial dipoles?
- All the radial dipoles were one color, and the random dots were a different color





- green radial dots
- red noise dots

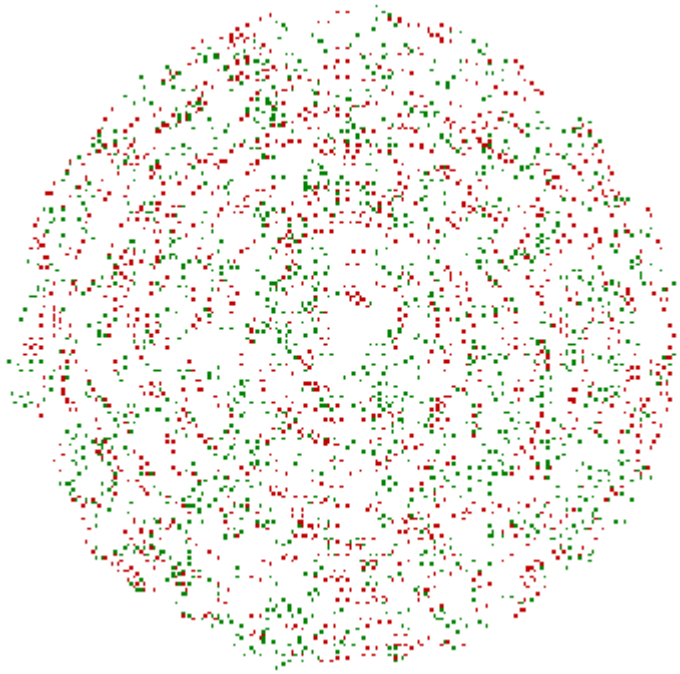
Pairing shape with color

- Takes time to determine which dots form the shape and which are random
- Consistent with attentional scrutiny being needed



Pairing shape with color

Signal dots one color, noise dots another color



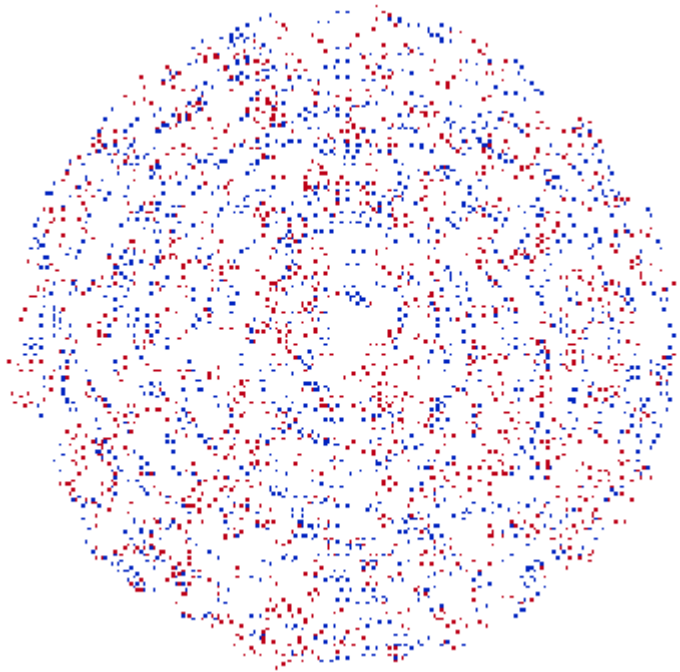
Quantifying binding cost

% of dots needed to perceive shape-color combination is:

- higher than needed for shape alone
- higher than needed for color alone
- e.g., 10% to perceive shape and 20% to perceive shape-color combo ¹

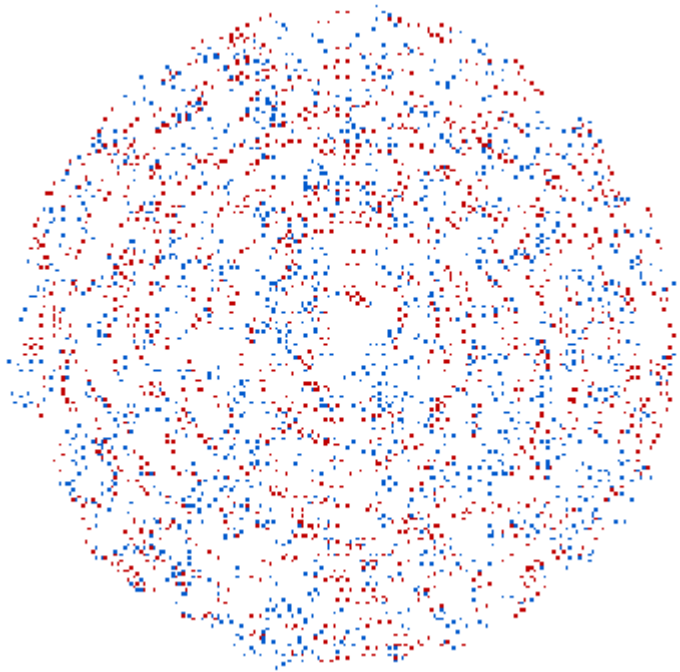
Pairing shape with color

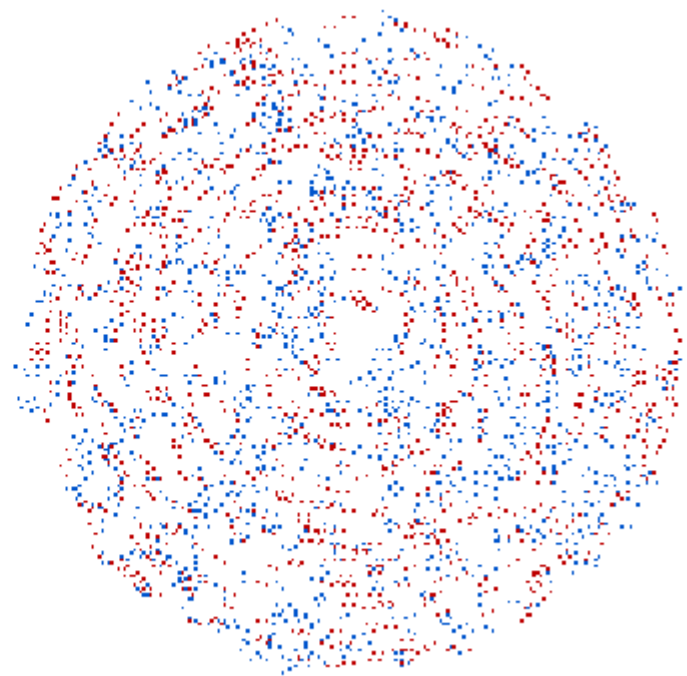
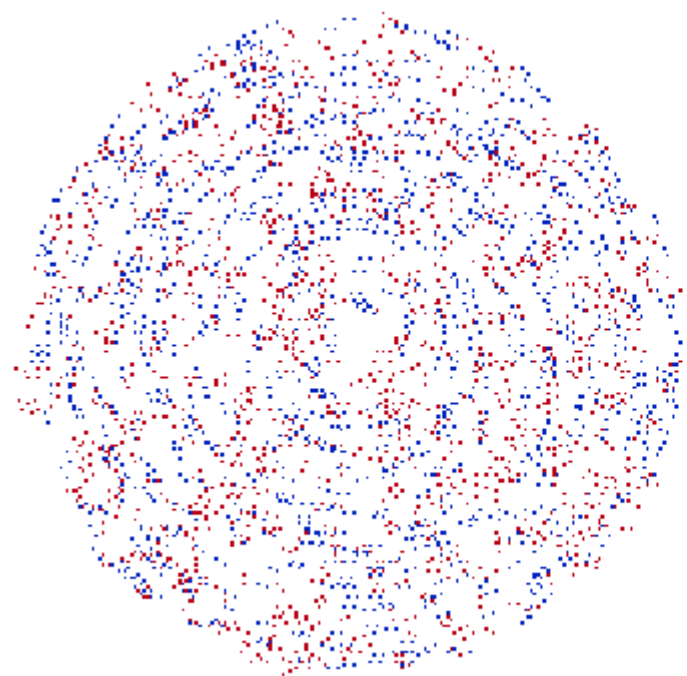
Signal dots are one color, noise dots are another color



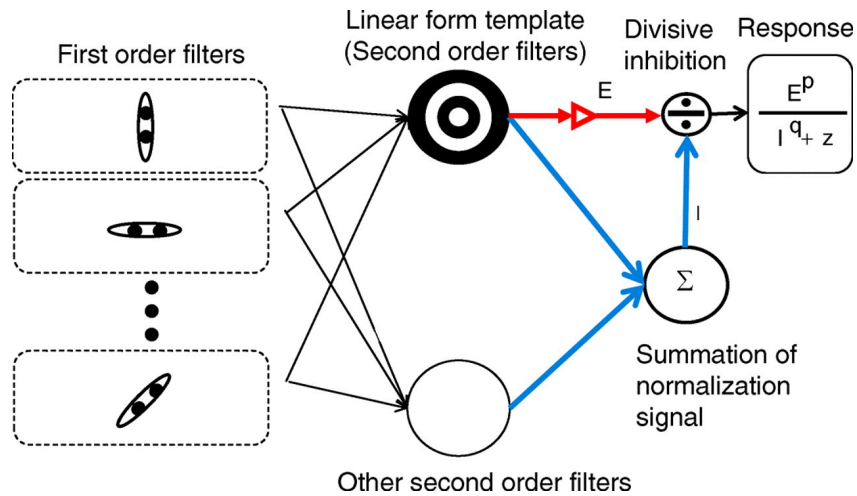
Pairing shape with color

Signal dots are one color, noise dots are another color

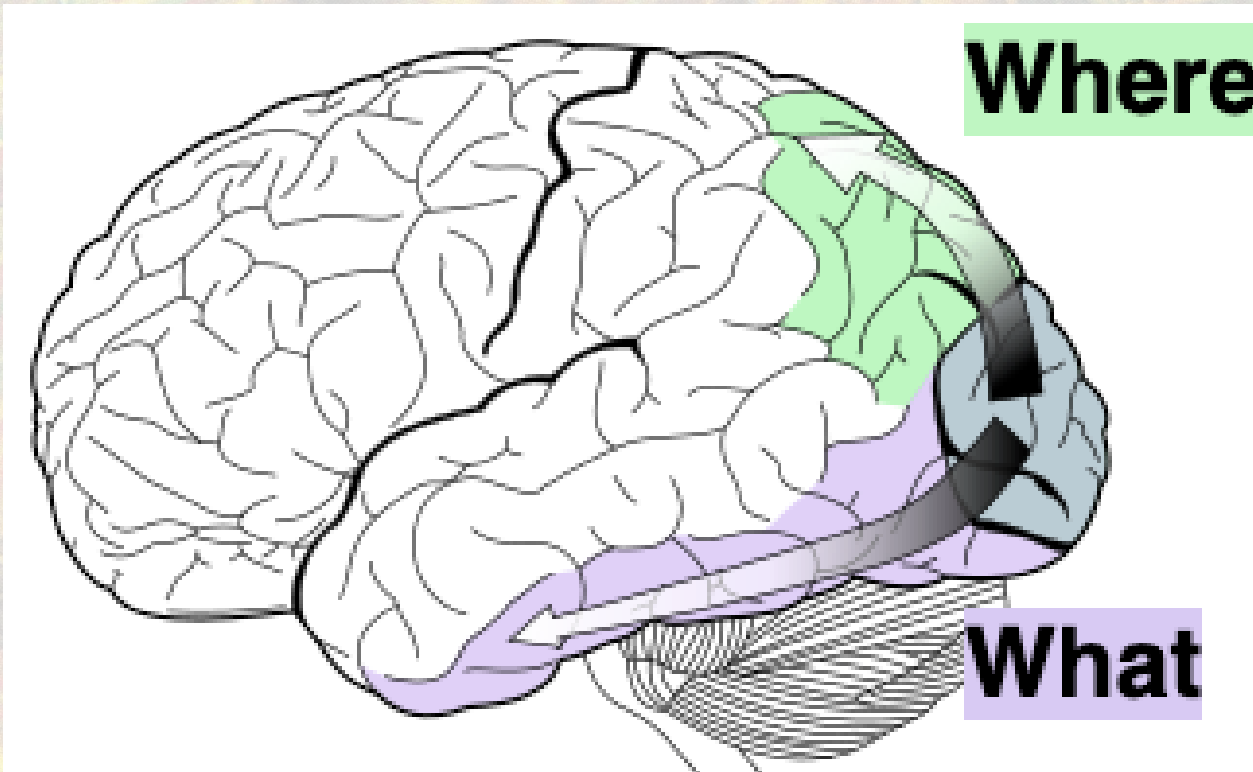




Shape extractor does not preserve color



- Orientations get bound together but not with color.



Shape take-aways

- Binding of individual orientations into single global shape is fast and may occur before bottleneck
- Color is processed independently, even though all the signal dots are that color

Feature pairing

- Cost to pairing different feature dimensions
 - Color with motion
 - Color with shape
- Need to bring together separate neural populations
 - Might only occur post-bottleneck